

Technology Stack

Requirements Specification - Rising Waters

1. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent (minimum); Intel i5 or above (recommended for model training).
- RAM: 4 GB (minimum); 8 GB (recommended for running local notebooks and training pipelines).
- Storage: 2 GB free disk space (to house datasets, virtual environments, and source code).
- Network: Active internet connection for Vercel/GitHub deployment and initial library installations.

Software Requirements

- Operating System: Windows 10/11, macOS, or Linux.
- Development Environment: VS Code or PyCharm.
- Python Runtime: Python 3.8 to 3.12.
- Web Framework: Flask (WSGI web micro-framework).
- Web Libraries: Bootstrap 5, Bootstrap Icons, FontAwesome.
- ML Libraries (Local Development): Scikit-learn, XGBoost, Pandas, Numpy, Openpyxl.

2. Data Flow Diagrams (DFD)

Level 0 DFD: Context Diagram

```
graph TD
    User([Meteorologist / Coordinator]) -->|Input Weather Parameters| System[["Rising Waters Flood Hazard Forecasting System"]]
    System -->|Flood Warning / Safe Status| User
    System -->|Confidence Score %| User
```

Level 1 DFD: System Architecture

```
graph TD
    User([User]) -->|POST Web Form| App[["Flask Server app.py"]]
    App -->|Input Sanitization| Val[["Validation Engine"]]
    Val -->|If Errors| Flash[["Flashed Error Message"]]
    Flash -->|Redirect| User
    Val -->|If Valid| Scale[["Manual Scaling scaling_loop"]]
    Scale -->|Z-Score Features| Model[["Decision Tree Classifier"]]
    Model -->|Binary Class| Predict[["Prediction Result"]]
    Model -->|Sigmoid Confidence| Prob[["Confidence Score %"]]
    Predict & Prob -->|Render Outcome| Result[["chance / no_chance Card"]]
    Result -->|HTML Display| User
```